Trace norm regularization for multi-task learning with scarce data

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Introduction: multi-task/meta learning

Goal: learn high dimensional tasks with limited data for each of them

How? leverage parameters structure among tasks (eg shared representation)
$\rightarrow$ learn tasks collectively

Motivations:

- learning personalised models (web search, spam filter, medical diagnoses. . .)
- few shot supervised learning
- few shot reinforcement learning

This talk: learn linear regression tasks in high dimension
when regression parameters share a low dimensional space

Setting

T tasks, each with m observation samples $(x_i^t, y_i^t) \in \mathbb{R}^d \times \mathbb{R}$

Linear model $\quad y_i^t = \langle M^{*\,(t)}, x_i^t \rangle + \varepsilon_i^t \qquad$ with $M^* \in \mathbb{R}^{d \times T}$

Low dimensional structure $\quad \mathrm{rank}(M^*) = r \ll d$

Random design

x_i^t, ε_i^t are independent centered subgaussian random variables (ε_i^t is σ-subgaussian)

Isotropic features: $\mathbb{E}\left[(x_i^t)^\top x_i^t\right] = I_d$

Task diversity $\quad \max_{t \in [T]} \|M^{*\,(t)}\|_2 \leq C$

Goal: estimate parameters matrix M^* from observations

Problem: rank constrained ERM cannot be computed efficiently

$$\underset{\substack{M \in \mathbb{R}^{d \times T} \\ \mathrm{rank}(M) \leq r}}{\arg\min} \frac{1}{mT} \sum_{(i,t) \in [m] \times [T]} \left(y_i^t - \langle M^{(t)}, x_i^t \rangle \right)^2$$

Classical methods

Burer-Monteiro factorisation

$$\underset{\substack{U \in \mathbb{R}^{d \times r} \\ V \in \mathbb{R}^{T \times r}}}{\arg\min} \frac{1}{mT} \sum_{(i,t) \in [m] \times [T]} \left(y_i^t - \langle (UV^\top)^{(t)}, x_i^t \rangle \right)^2 .$$

Equivalent problem, but non-convex
Bounds by Tripuraneni et al. (2021), but require $m \gtrsim \log T$

Method of Moments (Tripuraneni et al., 2021)

$\rightarrow$ strong assumption on features distribution

Trace norm regularisation

$$\underset{M \in \mathbb{R}^{d \times T}}{\arg\min} \frac{1}{mT} \sum_{(i,t) \in [m] \times [T]} \left(y_i^t - \langle M^{(t)}, x_i^t \rangle \right)^2 + \lambda \|M\|_* .$$

Trace norm convexifies the rank
Rohde and Tsybakov (2011) $\rightarrow$ tight error bound when $m \gtrsim d$
Our work: first general error bound when $m < d$

Main result

$$\hat{M} = \arg\min_{M \in \mathcal{W}} \frac{1}{mT} \sum_{(i,t) \in [m] \times [T]} (y_i^t - \langle x_i^t, M^{(t)} \rangle)^2 + \lambda \|M\|_*$$

where $\mathcal{W} = \left\{ M \in \mathbb{R}^{d \times T} \mid \max_{t \in [T]} \|M^{(t)}\|^2 \leq C \right\}$

Theorem

For $\lambda = \tau \approx \frac{\sigma}{\sqrt{mT}}$, with high probability

$$\|\hat{M} - M^*\|_F \lesssim \sigma \sqrt{\frac{rT}{m}} + \sqrt{\frac{rdT}{m^2}}$$

Estimator	**Error bound** $\|\hat{M} - M^*\|_F$	**Samples per task**
Single task	$\sigma \sqrt{\frac{dT}{m}}$	$\Omega(1)$
Burer-Monteiro factorization	$\sigma \sqrt{\frac{rT}{m}}$	$\Omega(r^4 \log(T))$
Method of Moments	$\sigma \sqrt{r \frac{T}{m}} + r \sqrt{\frac{d}{m}}$	$\Omega(r \log(r))$
Trace norm regularization	$\sigma \sqrt{\frac{rT}{m}}$	$\Omega(d)$
Trace norm regularization	$\sigma \sqrt{\frac{rT}{m}} + \sqrt{\frac{rdT}{m^2}}$	$\Omega(1)$

Sketch of proof

$$\mathcal{L}: \begin{array}{l} \mathbb{R}^{d\times T} \to \mathbb{R}^{m\times T} \\ M \mapsto \frac{1}{\sqrt{mT}}(\langle x_i^t, M^{(t)}\rangle)_{i,t} \end{array}$$

Rohde and Tsybakov (2011) use restricted isometry condition:
$\|\mathcal{L}(\Delta)\|_F^2 \gtrsim \frac{1}{T}\|\Delta\|_F^2$ for all $\Delta \in \mathbb{R}^{d\times T}$ $\to$ requires $m \geq d$

For $m < d$, restricted strong convexity condition.
For $\hat{\Delta} := \hat{M} - M^*$,

$$\overbrace{\frac{1}{T}\|\hat{\Delta}\|_F^2 - \frac{rd}{m^2}\max_{t\in[T]}\|\hat{\Delta}^{(t)}\|_2^2}^{\text{RSC condition}} \lesssim \|\mathcal{L}(\hat{\Delta})\|^2 \overbrace{\lesssim \frac{\sigma}{\sqrt{mT}}\|\hat{\Delta}\|_*}^{\text{effective noise level}}$$
$$\lesssim \frac{\sigma}{\sqrt{mT}}\sqrt{r}\|\hat{\Delta}\|_F$$

$\square$

Tightness of result

$$\|\hat{M} - M^*\|_F \lesssim \overbrace{\sigma\sqrt{\frac{rT}{m}}}^{\text{lower bound}} + \sqrt{\frac{rT}{m}}\,\overbrace{\sqrt{\frac{d}{m}}}^{\text{analysis artefact?}}$$

Belief: additional $\sqrt{\frac{d}{m}}$ term = artefact from analysis

- comes from RSC condition

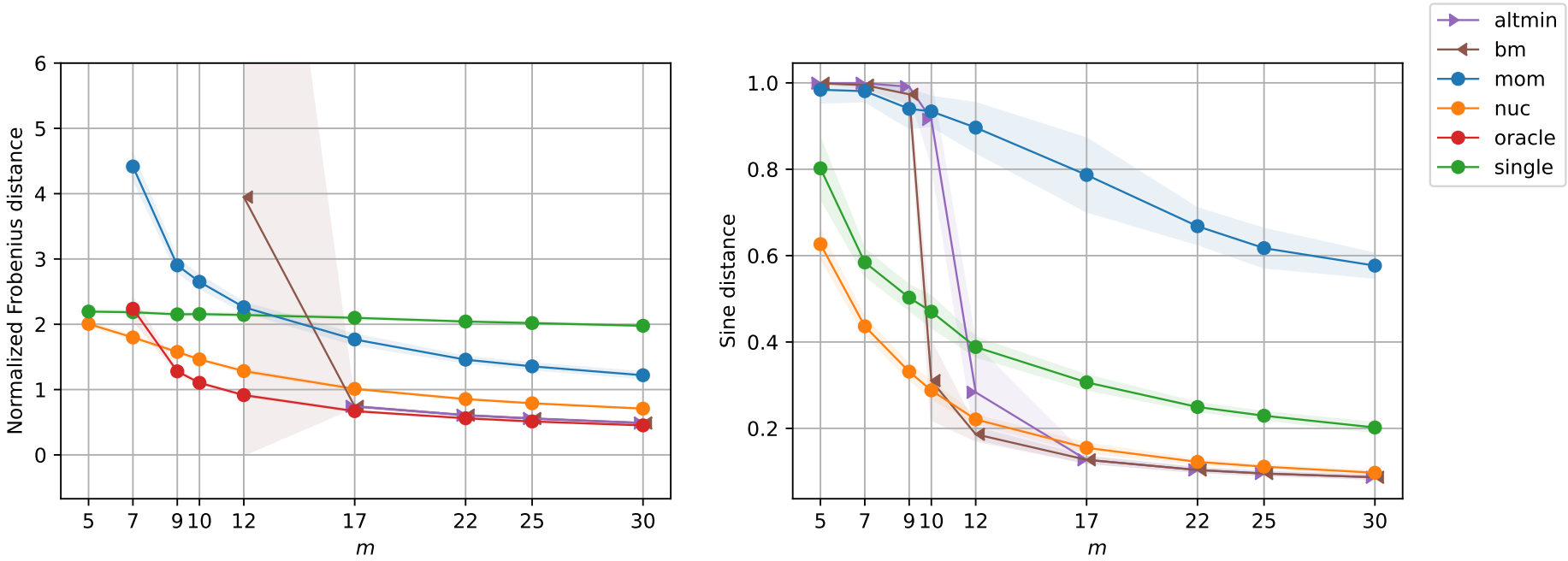
$$\|\mathcal{L}(\Delta)\|^2 \gtrsim \frac{1}{T}\|\Delta\|_F^2 - \frac{r}{m}\frac{d}{m}\max_{t\in[T]}\|\Delta^{(t)}\|_2^2 \qquad \text{for all } \Delta \in \mathcal{C} \qquad \text{(RSC)}$$

- covering arguments are not tight enough
- which analytical tools?

Trace norm regularisation is better than single task learning by a factor $\sqrt{\frac{r}{m}}$
($\sqrt{\frac{r}{d}}$ without artefact term)

Simulations

Left plot: estimation error in matrix norm
Right plot: principal angle between estimated and true low dimensional subspace



Evolution of the estimation error with the task size m ($T = 800, d = 100, r = 5$)

Trace norm regularisation is

- better for small task sizes m
- comparable to Burer Monteiro factorisation for intermediate task sizes

Conclusion

Summary

- First multi-task error bound with very limited data per task
- Additional meta-learning bound and simulations in the paper
- Illustration of failure of MoM in general settings in the paper

Open directions

- Is a tighter error bound possible?
- What about other methods in this scarce data regime?

Thank you for your attention!